EN 3160 - Assignment 3

Neural Networks

Index No. - 230013A

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Github - https://github.com/dilith813/en3160-a03

Question 1

In [13]:
```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt

# 1. Dataloading
transform = transforms.Compose(
    [transforms.ToTensor(),
     transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5))])

batch_size = 50

trainset = torchvision.datasets.CIFAR10(root='../data', train=True,
                                        download=True, transform=transform)
trainloader = torch.utils.data.DataLoader(trainset, batch_size=batch_size,
                                          shuffle=True, num_workers=2)
testset = torchvision.datasets.CIFAR10(root='../data', train=False,
                                       download=True, transform=transform)
testloader = torch.utils.data.DataLoader(testset, batch_size=batch_size,
                                         shuffle=False, num_workers=2)

classes = ('plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck')
```

First, the CIFAR-10 dataset was loaded. It has 50 000 training and 10 000 test colour images of 32x32 pixels, belonging to 10 classes. They were preprocessed, being converted into tensors and normalized to [-1,1]. Each image is flattened into a vector of 3072 (32x32x3) values. Since the images are processed in batches of size 50, an epoch would have 1000 weight updates.

In [14]:
```
# 2. Define Network Parameters
Din = 3 * 32 * 32      # Input size (flattened CIFAR-10 image size)
K = 10                 # Output size (number of classes in CIFAR-10)
std = 1e-2             # Value was changed from 1e-5 (to better suit two layers)

# Define Middle Layer Parameters (Q1 (a))
H = 100                                 # number of nodes in the middle layer
torch.manual_seed(0)                    # added for reproducibility
w1 = torch.randn(Din, H) * std          # layer 1: input -> middle
b1 = torch.zeros(H)
w2 = torch.randn(H, K) * std            # layer 2: middle -> output
b2 = torch.zeros(K)

# Hyperparameters
iterations = 10     # 10 epochs
lr = 0.05         # Learning rate
lr_decay = 0.9      # Learning rate decay
```

```
reg = 0              # Regularization
loss_history = []
```

Q1(a): Network architecture

The single dense layer of Listing 1 was replaced by two layers:

$$z_1 = xW_1 + b_1, \quad h = \sigma(z_1), \quad \hat{y} = hW_2 + b_2$$

with $x \in \mathbb{R}^{3072}$, $h \in \mathbb{R}^{100}$ and $\hat{y} \in \mathbb{R}^{10}$, where $\sigma(z) = 1/(1 + e^{-z})$. This gives 3072·100 + 100 + 100·10 + 10 = 308,310 trainable parameters. Weights were initialised from a normal distribution with std 0.01 and biases to zero.

Random weights break the symmetry between hidden nodes, which would otherwise all receive identical gradients.

The sigmoid adds non-linearity. Without it, the two stacked linear layers would collapse into one linear map.

Hyperparameters. Batch size 50, 10 epochs, learning rate 0.05 decayed by 0.9 after each epoch, no regularization (reg = 0). The listing's values (std = 1e-5, lr = 2e-6) were designed for a single linear layer. With lr = 2e-6 the loss stayed at about 2.3036 for all 10 epochs (ln 10 ≈ 2.3026, the loss of a uniform guess), so the network did not learn (the 2e-6 run used std=0.01 as well). The updates were too small. lr = 0.05 was chosen because it trains smoothly; it was not tuned further.

In [15]:
```
# 3. Training Loop
for t in range(iterations):
    running_loss = 0.0
    for i, data in enumerate(trainloader, 0):
        # Get inputs and labels
        inputs, labels = data
        Ntr = inputs.shape[0]                                          # Batch size
        x_train = inputs.view(Ntr, -1)                                 # Flatten input to (N
tr, Din)
        y_train_onehot = nn.functional.one_hot(labels, K).float()      # Convert labels to o
ne-hot

        # Forward Pass with 2 layers
        z1 = x_train.mm(w1) + b1                                       # middle layer pre-ac
tivation
        h = 1.0 / (1.0 + torch.exp(-z1))                               # sigmoid activation
        y_pred = h.mm(w2) + b2                                         # output layer scores
(logits)

        # Calculating Cross-Entropy Loss
        z = y_pred - y_pred.max(dim=1, keepdim=True).values
        e = torch.exp(z)
        p = e / e.sum(dim=1, keepdim=True)                             # softmax probabiliti
es
        loss = -torch.log(p[torch.arange(Ntr), labels]).mean() \
               + reg * (torch.sum(w1 ** 2) + torch.sum(w2 ** 2))       # cross-entropy + reg
ularization

        loss_history.append(loss.item())
        running_loss += loss.item()

        # Backpropagation
        dy_pred = (p - y_train_onehot) / Ntr                           # softmax + cross-ent
ropy gradient

        dw2 = h.t().mm(dy_pred) + reg * w2
        db2 = dy_pred.sum(dim=0)
        dh = dy_pred.mm(w2.t())                                        # push gradient back
```

```
through w2
        dz1 = dh * h * (1 - h)                                          # sigmoid derivative
        dw1 = x_train.t().mm(dz1) + reg * w1
        db1 = dz1.sum(dim=0)

        # Parameter update
        w2 -= lr * dw2
        b2 -= lr * db2
        w1 -= lr * dw1
        b1 -= lr * db1

    # Print loss for every epoch
    if t % 1 == 0:
        print(f"Epoch {t + 1}/{iterations}, Loss: {running_loss / len(trainloader)}")

    # Learning rate decay
    lr *= lr_decay

print(f"Loss before any learning: {loss_history[0]}")
```

```
Epoch 1/10, Loss: 1.983628683924675
Epoch 2/10, Loss: 1.7916365011930466
Epoch 3/10, Loss: 1.7248878792524338
Epoch 4/10, Loss: 1.6850644743442535
Epoch 5/10, Loss: 1.6558336367607116
Epoch 6/10, Loss: 1.63202667927742
Epoch 7/10, Loss: 1.612223219871521
Epoch 8/10, Loss: 1.5950423702001573
Epoch 9/10, Loss: 1.5796113493442536
Epoch 10/10, Loss: 1.5665409650802613
Loss before any learning: 2.3027901649475098
```

Q1(a): Backpropagation through the two layers

The gradient is propagated backwards with the chain rule:

$$dW_2 = h^\top d\hat{y}, \quad db_2 = \sum d\hat{y}, \quad dh = d\hat{y}\, W_2^\top$$

$$dz_1 = dh \odot h \odot (1 - h), \quad dW_1 = x^\top dz_1, \quad db_1 = \sum dz_1$$

The factor $h(1-h)$ is the sigmoid derivative $\sigma'(z) = \sigma(z)(1-\sigma(z))$. Each backward line mirrors one forward line: a matrix product aW gives $dW = a^\top dz$ and $da = dz\, W^\top$. Parameters are then updated by plain gradient descent, $W \leftarrow W - \eta\, dW$.

Q1(b): Cross-entropy loss

The scores are converted to probabilities with softmax,

$$p_k = \frac{e^{\hat{y}_k}}{\sum_j e^{\hat{y}_j}}$$

The largest score is subtracted first so that the exponentials don't overflow. This does not affect the result.

Cross-Entropy loss only looks at the probability given to the correct class and penalizes it for being small.

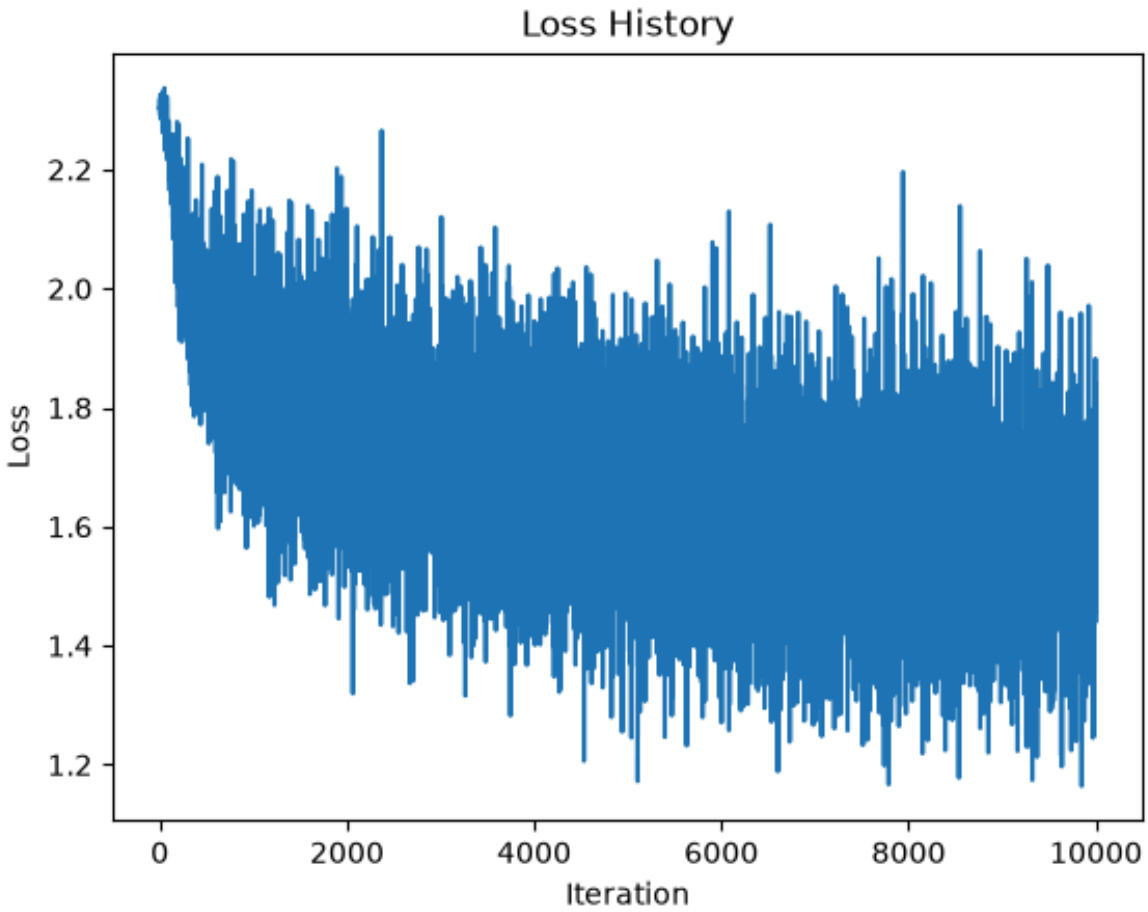
$$L = -\frac{1}{N} \sum_n \ln p_{n,\text{correct}}$$

(For example, if the correct class gets probability 0.9 the loss for that image is 0.11; if it gets 0.1, the loss would be 2.30. When training starts, all 10 classes start at about 10%

probability. So, the loss would start at ≈ 2.30, as observed above.)

Combining softmax and cross-entropy gives a gradient of (predicted probabilities - one-hot labels). In code: 'dy_pred = (p - y_onehot) / N'. This replaces the MSE gradient of the listing, $(2/N)(\hat{y} - y)$.

In [16]:
```
# 4. Plotting the Loss History
plt.plot(loss_history)
plt.title("Loss History")
plt.xlabel("Iteration")
plt.ylabel("Loss")
plt.show()
```

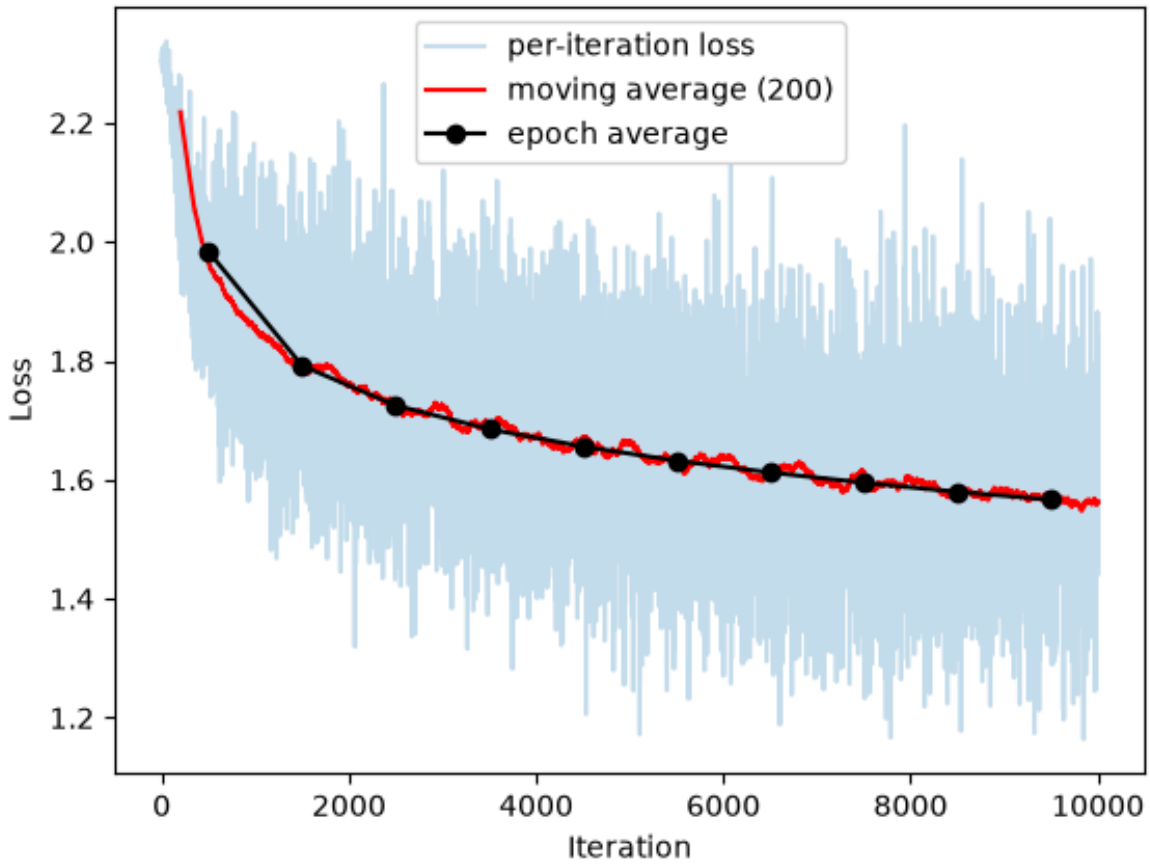


In [19]:
```
# Smoothed Loss History (For better visualization of trend)
import numpy as np

lh = np.array(loss_history)
win = 200                                          # moving-average window (iterations)
smooth = np.convolve(lh, np.ones(win) / win, mode='valid')
epoch_avg = lh.reshape(10, -1).mean(axis=1)        # 1,000 updates per epoch

plt.plot(lh, alpha=0.25, label='per-iteration loss')
plt.plot(np.arange(win - 1, len(lh)), smooth, 'r', label=f'moving average ({win})')
plt.plot(np.arange(10) * 1000 + 500, epoch_avg, 'ko-', label='epoch average')
plt.xlabel('Iteration'); plt.ylabel('Loss'); plt.legend(); plt.show()

print("first 100 iterations, mean loss:", lh[:100].mean())
print("last 1000 iterations, mean loss:", lh[-1000:].mean())
print("per-iteration std in last 1000:", lh[-1000:].std())
```




```
first 100 iterations, mean loss: 2.280981769561768
last 1000 iterations, mean loss: 1.5665409650802613
per-iteration std in last 1000: 0.1362109500779254
```

The loss starts at about 2.3 and drops quickly over the first few hundred iterations, then keeps falling slowly. The raw curve is very noisy because each point is the loss on only 50 images, and some batches are easier than others. The moving average shows the trend underneath, and the epoch averages fall from 1.98 to 1.57. The improvement per epoch shrinks (about 0.19 after the first epoch, about 0.01 at the end), so training is slowing down, helped by the decaying learning rate, but it has not fully converged within the 10 epochs.

In [17]:
```python
# 5. Calculate Accuracy on Training Set
correct_train = 0
total_train = 0
with torch.no_grad():
    for data in trainloader:
        inputs, labels = data
        Ntr = inputs.shape[0]
        x_train = inputs.view(Ntr, -1)
        y_train_onehot = nn.functional.one_hot(labels, K).float()

        # Forward pass through both layers
        h = 1.0 / (1.0 + torch.exp(-(x_train.mm(w1) + b1)))
        y_train_pred = h.mm(w2) + b2

        predicted_train = torch.argmax(y_train_pred, dim=1)
        total_train += labels.size(0)
        correct_train += (predicted_train == labels).sum().item()
train_acc = 100 * correct_train / total_train
# Calculate Training Accuracy (Q1 (c))
print(f"Training accuracy: {train_acc:.2f}%")
```

```
Training accuracy: 46.48%
```

In [18]:
```python
# 6. Calculate Accuracy on Test Set
correct_test = 0
total_test = 0
```

```
with torch.no_grad():
    for data in testloader:
        inputs, labels = data
        Nte = inputs.shape[0]
        x_test = inputs.view(Nte, -1)
        y_test_onehot = nn.functional.one_hot(labels, K).float()

        # Forward pass through both layers
        h = 1.0 / (1.0 + torch.exp(-(x_test.mm(w1) + b1)))
        y_test_pred = h.mm(w2) + b2

        predicted_test = torch.argmax(y_test_pred, dim=1)
        total_test += labels.size(0)
        correct_test += (predicted_test == labels).sum().item()
test_acc = 100 * correct_test / total_test
# Calculate Test Accuracy (Q1 (c))
print(f"Test accuracy: {test_acc:.2f}%")
```

```
Test accuracy: 44.73%
```

Q1(c): Results after 10 epochs

	Accuracy
Training	46.48%
Test	44.73%

Both are well above the 10% chance level, so the network learned. The gap is only about 1.75 points, so there is no sign of overfitting. The accuracy is low because flattening the image into a vector throws away its 2-D structure. So, the network cannot detect local patterns such as edges or shapes, and every pixel is treated as an unrelated input.

Question 2

In [20]:

```
import torch.nn.functional as F
import time
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Device:", device)

# 0.1307 and 0.3081 are the mean and std of all MNIST pixels
mnist_tf = transforms.Compose([transforms.ToTensor(), transforms.Normalize((0.1307,), (0.308
1,))])
mnist_train = torchvision.datasets.MNIST('../data', train=True,  download=True, transform=mni
st_tf)
mnist_test  = torchvision.datasets.MNIST('../data', train=False, download=True, transform=mni
st_tf)

# num_workers=2 as in the assignment's listing; set both to 0 if the cell hangs on Windows
mnist_trainloader = torch.utils.data.DataLoader(mnist_train, batch_size=64,  shuffle=True,  n
um_workers=2)
mnist_testloader  = torch.utils.data.DataLoader(mnist_test,  batch_size=256, shuffle=False, n
um_workers=2)
print(len(mnist_train), len(mnist_test))
```

```
Device: cuda
60000 10000
```

In [21]:

```
# Q2(a) - LeNet-5
class LeNet5(nn.Module):
    def __init__(self):
        super().__init__()
        self.c1 = nn.Conv2d(1, 6, 5, padding=2)    # 28x28 -> 28x28 (padding=2 mimics LeNet's
32x32 input)
        self.c2 = nn.Conv2d(6, 16, 5)              # 14x14 -> 10x10
```

```
        self.f1 = nn.Linear(16 * 5 * 5, 120)
        self.f2 = nn.Linear(120, 84)
        self.f3 = nn.Linear(84, 10)
    def forward(self, x):
        x = F.avg_pool2d(torch.tanh(self.c1(x)), 2)    # -> 6 x 14 x 14
        x = F.avg_pool2d(torch.tanh(self.c2(x)), 2)    # -> 16 x 5 x 5
        x = x.flatten(1)
        x = torch.tanh(self.f1(x))
        x = torch.tanh(self.f2(x))
        return self.f3(x)                              # 10 raw scores

# Q2(b) - Modern small CNN: 3 conv layers + batch norm + dropout
class SmallCNN(nn.Module):
    def __init__(self):
        super().__init__()
        def block(i, o):                               # conv -> batch norm -> ReLU -> downsa
mple
            return nn.Sequential(nn.Conv2d(i, o, 3, padding=1), nn.BatchNorm2d(o),
                                 nn.ReLU(), nn.MaxPool2d(2))
        self.features = nn.Sequential(block(1, 32), block(32, 64), block(64, 128))  # 28 -> 1
4 -> 7 -> 3
        self.drop = nn.Dropout(0.5)
        self.fc = nn.Linear(128 * 3 * 3, 10)
    def forward(self, x):
        return self.fc(self.drop(self.features(x).flatten(1)))

print("LeNet-5 parameters:", sum(p.numel() for p in LeNet5().parameters()))
print("SmallCNN parameters:", sum(p.numel() for p in SmallCNN().parameters()))
```

```
LeNet-5 parameters: 61706
SmallCNN parameters: 104650
```

Q2(a) and (b): Models

LeNet-5 (61,706 parameters): two 5×5 convolution layers (6 then 16 filters), each followed by tanh and average pooling, then three fully connected layers (120, 84, 10). Tanh and average pooling was used as per the original paper. Padding of 2 in the first layer lets the 28×28 MNIST images behave like LeNet's original 32×32 input.

Small CNN (104,650 parameters): three 3×3 convolution blocks (32, 64, 128 filters), each with batch normalization, ReLU and 2×2 max pooling, then dropout (p = 0.5) and one fully connected layer. Batch normalization rescales each layer's outputs, which usually makes training faster and more stable. Dropout randomly switches off half of the features during training, which discourages the network from relying on any single one.

Training conditions (identical for both): MNIST, cross-entropy loss, Adam with learning rate 0.001, batch size 64, 10 epochs, same random seed. Train and test metrics were computed after every epoch with dropout off.

In [22]:

```
def evaluate(model, loader):
    model.eval()                                       # switches off dropout, batch norm use
s stored statistics
    loss_sum, correct, total, preds, truth = 0.0, 0, 0, [], []
    with torch.no_grad():
        for x, y in loader:
            x, y = x.to(device), y.to(device)
            out = model(x)
            loss_sum += F.cross_entropy(out, y, reduction='sum').item()
            pred = out.argmax(1)
            correct += (pred == y).sum().item(); total += y.size(0)
            preds.append(pred.cpu()); truth.append(y.cpu())
    return loss_sum / total, 100 * correct / total, torch.cat(preds).numpy(), torch.cat(trut
h).numpy()

def train_model(model, epochs=10, lr=1e-3):
    model.to(device)
    opt = torch.optim.Adam(model.parameters(), lr=lr)
```

```
    hist = dict(train_loss=[], train_acc=[], test_loss=[], test_acc=[])
    for ep in range(epochs):
        t0 = time.time()
        model.train()                                   # dropout on, batch norm uses batch st
atistics
        for x, y in mnist_trainloader:
            x, y = x.to(device), y.to(device)
            opt.zero_grad()
            F.cross_entropy(model(x), y).backward()
            opt.step()
        # Q2(c) - log loss and accuracy on both sets after every epoch (both in eval mode)
        trl, tra, _, _ = evaluate(model, mnist_trainloader)
        tel, tea, _, _ = evaluate(model, mnist_testloader)
        for key, val in zip(hist, (trl, tra, tel, tea)):
            hist[key].append(val)
        print(f"Epoch {ep+1:2d}/{epochs}  train loss {trl:.4f} acc {tra:.2f}%  |  "
              f"test loss {tel:.4f} acc {tea:.2f}%  ({time.time()-t0:.0f}s)")
    return hist
```

In [23]:
```
# Q2(a) - Train LeNet-5
torch.manual_seed(0)
lenet = LeNet5()
hist_lenet = train_model(lenet)

# Q2(b) Train SmallCNN, same optimiser, epochs, batch size and seed
torch.manual_seed(0)
small = SmallCNN()
hist_small = train_model(small)
```

```
Epoch  1/10  train loss 0.0866 acc 97.47%  |  test loss 0.0880 acc 97.38%  (36s)
Epoch  2/10  train loss 0.0505 acc 98.47%  |  test loss 0.0584 acc 98.25%  (33s)
Epoch  3/10  train loss 0.0399 acc 98.80%  |  test loss 0.0585 acc 98.28%  (33s)
Epoch  4/10  train loss 0.0313 acc 99.00%  |  test loss 0.0484 acc 98.47%  (33s)
Epoch  5/10  train loss 0.0236 acc 99.28%  |  test loss 0.0438 acc 98.62%  (30s)
Epoch  6/10  train loss 0.0191 acc 99.38%  |  test loss 0.0393 acc 98.63%  (60s)
Epoch  7/10  train loss 0.0194 acc 99.39%  |  test loss 0.0457 acc 98.59%  (55s)
Epoch  8/10  train loss 0.0170 acc 99.45%  |  test loss 0.0434 acc 98.75%  (57s)
Epoch  9/10  train loss 0.0105 acc 99.70%  |  test loss 0.0412 acc 98.62%  (54s)
Epoch 10/10  train loss 0.0130 acc 99.57%  |  test loss 0.0441 acc 98.72%  (55s)
Epoch  1/10  train loss 0.0525 acc 98.34%  |  test loss 0.0568 acc 98.33%  (56s)
Epoch  2/10  train loss 0.0275 acc 99.15%  |  test loss 0.0289 acc 99.06%  (59s)
Epoch  3/10  train loss 0.0195 acc 99.39%  |  test loss 0.0227 acc 99.23%  (68s)
Epoch  4/10  train loss 0.0140 acc 99.57%  |  test loss 0.0221 acc 99.22%  (37s)
Epoch  5/10  train loss 0.0156 acc 99.54%  |  test loss 0.0264 acc 99.22%  (36s)
Epoch  6/10  train loss 0.0135 acc 99.56%  |  test loss 0.0231 acc 99.33%  (35s)
Epoch  7/10  train loss 0.0141 acc 99.57%  |  test loss 0.0235 acc 99.25%  (39s)
Epoch  8/10  train loss 0.0101 acc 99.72%  |  test loss 0.0269 acc 99.16%  (37s)
Epoch  9/10  train loss 0.0067 acc 99.81%  |  test loss 0.0227 acc 99.35%  (36s)
Epoch 10/10  train loss 0.0061 acc 99.81%  |  test loss 0.0226 acc 99.26%  (36s)
```

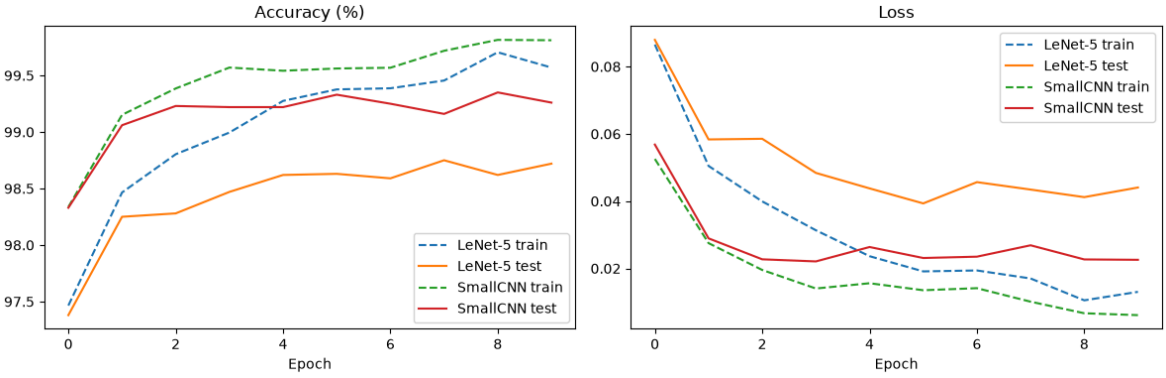
In [24]:
```
# Q2(a) and Q2(b) - Final results
results = {}
for name, m in [("LeNet-5", lenet), ("SmallCNN", small)]:
    trl, tra, _, _ = evaluate(m, mnist_trainloader)
    tel, tea, pred, truth = evaluate(m, mnist_testloader)
    results[name] = (pred, truth)
    print(f"{name:9s} train acc {tra:.2f}%   test acc {tea:.2f}%   test errors {(pred != trut
h).sum()} / {len(truth)}")

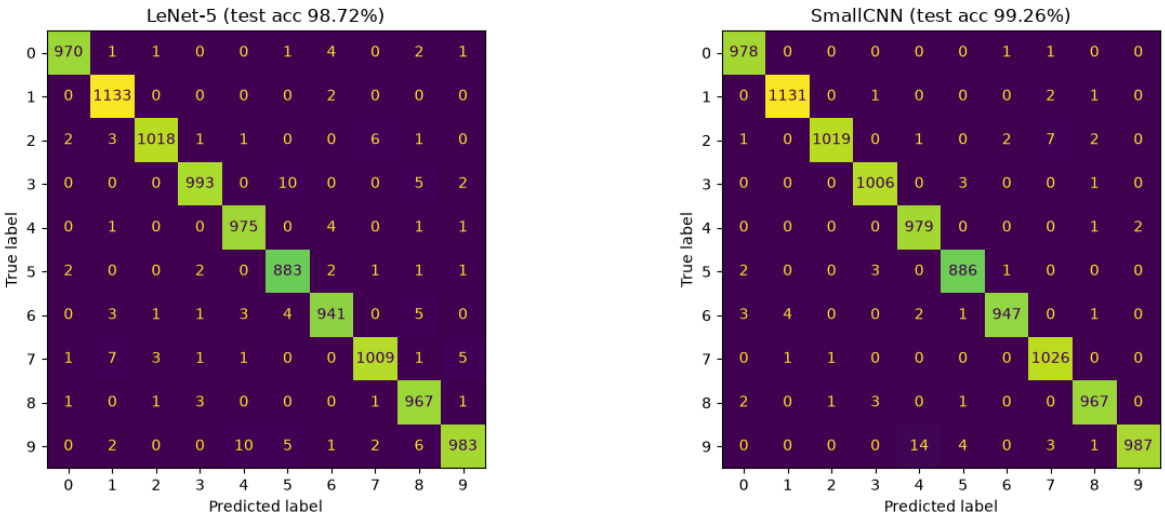
# Q2(c) - per-epoch accuracy and loss curves
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
for name, h in [("LeNet-5", hist_lenet), ("SmallCNN", hist_small)]:
    ax[0].plot(h['train_acc'], '--', label=f"{name} train"); ax[0].plot(h['test_acc'], label=
f"{name} test")
    ax[1].plot(h['train_loss'], '--', label=f"{name} train"); ax[1].plot(h['test_loss'], labe
l=f"{name} test")
ax[0].set_title("Accuracy (%)"); ax[1].set_title("Loss")
for a in ax: a.set_xlabel("Epoch"); a.legend()
plt.tight_layout(); plt.show()

# Q2(c) Confusion matrices on the test set
fig, ax = plt.subplots(1, 2, figsize=(13, 5))
for a, name in zip(ax, results):
    pred, truth = results[name]
    ConfusionMatrixDisplay(confusion_matrix(truth, pred)).plot(ax=a, colorbar=False)
    a.set_title(f"{name} (test acc {100*(pred == truth).mean():.2f}%)")
plt.tight_layout(); plt.show()
```

```python
# most frequent mistakes
for name, (pred, truth) in results.items():
    cm = confusion_matrix(truth, pred); np.fill_diagonal(cm, 0)
    top = np.dstack(np.unravel_index(np.argsort(-cm, axis=None)[:4], cm.shape))[0]
    print(name, "top confusions (true -> predicted: count):",
          [f"{t}->{p}: {cm[t, p]}" for t, p in top])
```

```
LeNet-5   train acc 99.57%   test acc 98.72%   test errors 128 / 10000
SmallCNN  train acc 99.81%   test acc 99.26%   test errors 74 / 10000
```






```
LeNet-5 top confusions (true -> predicted: count): ['3->5: 10', '9->4: 10', '7->1: 7', '2->7:
6']
SmallCNN top confusions (true -> predicted: count): ['9->4: 14', '2->7: 7', '9->5: 4', '6->1:
4']
```

Q2(a) and (b): Results after 10 epochs

Model	Train acc	Test acc	Test errors (of 10,000)	Parameters
LeNet-5	99.57%	98.72%	128	61,706
Small CNN	99.81%	99.26%	74	104,650

The small CNN is better, and the improvement is clearer as it makes about 42% fewer mistakes on the test set. However, there are two caveats. It also has about 1.7× more parameters, so the gain can't be credited only to batch normalization and dropout. And each result is a single run with one seed, so a difference of a few test images could change with another seed.

Q2(c): Training curves

Both models improve quickly in the first epochs. The small CNN is already at 98.33% test accuracy after one epoch, while LeNet-5 needs several epochs to pass 98.5%, which fits

the faster training that batch normalization usually gives. The small CNN's test accuracy stays above LeNet-5's at every epoch.

LeNet-5's training loss keeps falling, but its test loss stops improving after about epoch 6 (0.039) and then rises slightly, so it starts to overfit mildly. The small CNN's test loss stays roughly flat at about 0.022, and its train/test accuracy gap is smaller (0.55 vs 0.85 points), consistent with dropout limiting overfitting. The test curves are noisy from epoch to epoch, so small differences between neighbouring epochs shouldn't be over-interpreted.

Q2(c): Confusion matrices

Both matrices are strongly diagonal. The most common mistake for both models is a true 9 predicted as 4 (10 times for LeNet-5, 14 for the small CNN). The reverse (4 predicted as 9) is rare, so the confusion is one-directional. A likely reason for this is that handwritten 9s with a small or open loop look like a 4. LeNet-5 also confuses [3→5] (10) and [7→1] (7) noticeably, and the small CNN almost removes these (3 and 1). The pair 9→4 got slightly worse for the small CNN even though its total errors fell from 128 to 74, so the improvement is not uniform. The 2→7 confusion is essentially unchanged (6 and 7), which suggests those test digits are genuinely ambiguous.

Q2: Summary

The small CNN gives higher test accuracy (99.26% vs 98.72%), converges faster and overfits less, at the cost of more parameters. The two experiments (Q1 and Q2) use different datasets (CIFAR-10 and MNIST), so their accuracies cannot be directly compared. What the two experiments do show is that convolutional layers, which look at local patterns and share weights across positions, suit images better than a network that flattens the image into an unrelated vector.

Question 3

In [26]:
```
import os, copy, urllib.request, zipfile
from torchvision import datasets, models
from torch.optim import lr_scheduler

# Download the hymenoptera dataset (ants vs bees); skipped if already present
hym_root = '../data/hymenoptera_data'
if not os.path.exists(hym_root):
    urllib.request.urlretrieve('https://download.pytorch.org/tutorial/hymenoptera_data.zip',
                               '../data/hymenoptera_data.zip')
    zipfile.ZipFile('../data/hymenoptera_data.zip').extractall('../data')

# Preprocessing: ImageNet mean/std, because ResNet18 was pretrained with these
imagenet_norm = transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
hym_tf = {
    'train': transforms.Compose([transforms.RandomResizedCrop(224), transforms.RandomHorizont
alFlip(),
                                 transforms.ToTensor(), imagenet_norm]),
    'val':   transforms.Compose([transforms.Resize(256), transforms.CenterCrop(224),
                                 transforms.ToTensor(), imagenet_norm])}
hym_ds = {s: datasets.ImageFolder(f'{hym_root}/{s}', hym_tf[s]) for s in hym_tf}
hym_dl = {s: torch.utils.data.DataLoader(hym_ds[s], batch_size=4, shuffle=(s == 'train'), num
_workers=2)
```

```
          for s in hym_tf}
print(hym_ds['train'].classes, {s: len(hym_ds[s]) for s in hym_ds})
```

```
['ants', 'bees'] {'train': 244, 'val': 153}
```

Q3: Data and approach

The hymenoptera dataset has 244 training and 153 validation images of ants and bees (2 classes). Images were resized to 224×224 and normalised with the ImageNet mean and std, since ResNet18 was pretrained on ImageNet-1K. Training images also get random crops and horizontal flips (augmentation), which helps because the dataset is so small.

Both experiments start from the same pretrained ResNet18 with its 1000-class output layer replaced by a 2-class layer. They use identical training settings: SGD with momentum 0.9, learning rate 0.001 reduced by a factor of 10 every 7 epochs, batch size 4, 25 epochs. The only difference is which weights are allowed to change.

In [27]:
```
# One training function for both (a) and (b), so conditions are identical
def train_hym(model, optimizer, scheduler, epochs=25):
    best_w, best_acc = copy.deepcopy(model.state_dict()), 0.0
    hist = {'train_acc': [], 'val_acc': [], 'train_loss': [], 'val_loss': [], 'time': []}
    for ep in range(epochs):
        t0 = time.time()
        for phase in ['train', 'val']:
            model.train(phase == 'train')              # train mode: batch norm uses batch st
atistics
            loss_sum, correct = 0.0, 0
            for x, y in hym_dl[phase]:
                x, y = x.to(device), y.to(device)
                optimizer.zero_grad()
                with torch.set_grad_enabled(phase == 'train'):   # no gradients needed for va
lidation
                    out = model(x)
                    loss = F.cross_entropy(out, y)
                    if phase == 'train':
                        loss.backward(); optimizer.step()
                loss_sum += loss.item() * x.size(0)
                correct += (out.argmax(1) == y).sum().item()
            if phase == 'train':
                scheduler.step()                       # lower the learning rate on a schedul
e
            n = len(hym_ds[phase])
            hist[phase + '_acc'].append(correct / n)
            hist[phase + '_loss'].append(loss_sum / n)
        hist['time'].append(time.time() - t0)
        if hist['val_acc'][-1] > best_acc:             # remember the best epoch (by validati
on accuracy)
            best_acc, best_w = hist['val_acc'][-1], copy.deepcopy(model.state_dict())
        print(f"Epoch {ep+1:2d}/{epochs}  train acc {hist['train_acc'][-1]:.3f}  "
              f"val acc {hist['val_acc'][-1]:.3f}  ({hist['time'][-1]:.0f}s)")
    model.load_state_dict(best_w)                      # go back to the best epoch's weights
    return model, hist, best_acc
```

In [28]:
```
# Fine-tuning: start from ImageNet weights, replace the last layer, train ALL layers
torch.manual_seed(0)
resnet_ft = models.resnet18(weights=models.ResNet18_Weights.IMAGENET1K_V1)
resnet_ft.fc = nn.Linear(resnet_ft.fc.in_features, 2)          # 1000 ImageNet classes -> 2
(ant, bee)
resnet_ft = resnet_ft.to(device)

opt_ft = optim.SGD(resnet_ft.parameters(), lr=0.001, momentum=0.9)
sched_ft = lr_scheduler.StepLR(opt_ft, step_size=7, gamma=0.1)       # lr x 0.1 every 7 epoch
s
resnet_ft, hist_ft, best_ft = train_hym(resnet_ft, opt_ft, sched_ft)
trainable_ft = sum(p.numel() for p in resnet_ft.parameters() if p.requires_grad)
print("Q3(a) fine-tuning: best val acc", best_ft, " trainable parameters", trainable_ft)
```

```
Epoch  1/25  train acc 0.730  val acc 0.889  (18s)
Epoch  2/25  train acc 0.742  val acc 0.902  (18s)
Epoch  3/25  train acc 0.828  val acc 0.908  (17s)
Epoch  4/25  train acc 0.832  val acc 0.935  (18s)
Epoch  5/25  train acc 0.869  val acc 0.876  (20s)
Epoch  6/25  train acc 0.795  val acc 0.876  (17s)
Epoch  7/25  train acc 0.836  val acc 0.745  (15s)
Epoch  8/25  train acc 0.877  val acc 0.843  (17s)
Epoch  9/25  train acc 0.865  val acc 0.876  (17s)
Epoch 10/25  train acc 0.877  val acc 0.882  (16s)
Epoch 11/25  train acc 0.910  val acc 0.908  (16s)
Epoch 12/25  train acc 0.828  val acc 0.908  (16s)
Epoch 13/25  train acc 0.844  val acc 0.856  (19s)
Epoch 14/25  train acc 0.840  val acc 0.922  (18s)
Epoch 15/25  train acc 0.828  val acc 0.908  (19s)
Epoch 16/25  train acc 0.885  val acc 0.928  (19s)
Epoch 17/25  train acc 0.869  val acc 0.902  (19s)
Epoch 18/25  train acc 0.852  val acc 0.928  (19s)
Epoch 19/25  train acc 0.926  val acc 0.928  (18s)
Epoch 20/25  train acc 0.840  val acc 0.895  (19s)
Epoch 21/25  train acc 0.889  val acc 0.922  (18s)
Epoch 22/25  train acc 0.906  val acc 0.922  (19s)
Epoch 23/25  train acc 0.906  val acc 0.928  (20s)
Epoch 24/25  train acc 0.873  val acc 0.922  (20s)
Epoch 25/25  train acc 0.914  val acc 0.908  (20s)
Q3(a) fine-tuning: best val acc 0.934640522875817  trainable parameters 11177538
```

Q3(a): Fine-tuning

All layers of the pretrained network, about 11.2 million parameters, are trained, starting from the ImageNet weights. The best validation accuracy was 93.46% (at epoch 4). Validation accuracy was somewhat noisy from epoch to epoch (for example a dip to 74.5% at epoch 7), which is typical when many pretrained weights are updated using only 244 images.

In [29]:

```
# Feature extractor: freeze the whole pretrained network, train only the new last layer
torch.manual_seed(0)
resnet_fe = models.resnet18(weights=models.ResNet18_Weights.IMAGENET1K_V1)
for p in resnet_fe.parameters():
    p.requires_grad = False
resnet_fe.fc = nn.Linear(resnet_fe.fc.in_features, 2)
resnet_fe = resnet_fe.to(device)

opt_fe = optim.SGD(resnet_fe.fc.parameters(), lr=0.001, momentum=0.9)
sched_fe = lr_scheduler.StepLR(opt_fe, step_size=7, gamma=0.1)
resnet_fe, hist_fe, best_fe = train_hym(resnet_fe, opt_fe, sched_fe)
trainable_fe = sum(p.numel() for p in resnet_fe.parameters() if p.requires_grad)
print("Q3(b) feature extractor: best val acc", best_fe, " trainable parameters", trainable_f
e)
```

```
Epoch  1/25  train acc 0.660  val acc 0.941  (15s)
Epoch  2/25  train acc 0.721  val acc 0.869  (16s)
Epoch  3/25  train acc 0.820  val acc 0.902  (16s)
Epoch  4/25  train acc 0.848  val acc 0.941  (16s)
Epoch  5/25  train acc 0.828  val acc 0.869  (19s)
Epoch  6/25  train acc 0.762  val acc 0.915  (19s)
Epoch  7/25  train acc 0.779  val acc 0.882  (18s)
Epoch  8/25  train acc 0.852  val acc 0.948  (19s)
Epoch  9/25  train acc 0.832  val acc 0.941  (17s)
Epoch 10/25  train acc 0.877  val acc 0.941  (17s)
Epoch 11/25  train acc 0.906  val acc 0.948  (18s)
Epoch 12/25  train acc 0.848  val acc 0.954  (18s)
Epoch 13/25  train acc 0.848  val acc 0.928  (17s)
Epoch 14/25  train acc 0.824  val acc 0.941  (18s)
Epoch 15/25  train acc 0.795  val acc 0.941  (15s)
Epoch 16/25  train acc 0.852  val acc 0.948  (14s)
Epoch 17/25  train acc 0.898  val acc 0.948  (15s)
Epoch 18/25  train acc 0.832  val acc 0.928  (13s)
Epoch 19/25  train acc 0.861  val acc 0.948  (14s)
Epoch 20/25  train acc 0.836  val acc 0.935  (14s)
Epoch 21/25  train acc 0.852  val acc 0.941  (13s)
Epoch 22/25  train acc 0.857  val acc 0.948  (14s)
Epoch 23/25  train acc 0.852  val acc 0.948  (15s)
Epoch 24/25  train acc 0.857  val acc 0.954  (13s)
Epoch 25/25  train acc 0.902  val acc 0.948  (14s)
Q3(b) feature extractor: best val acc 0.954248366013072  trainable parameters 1026
```

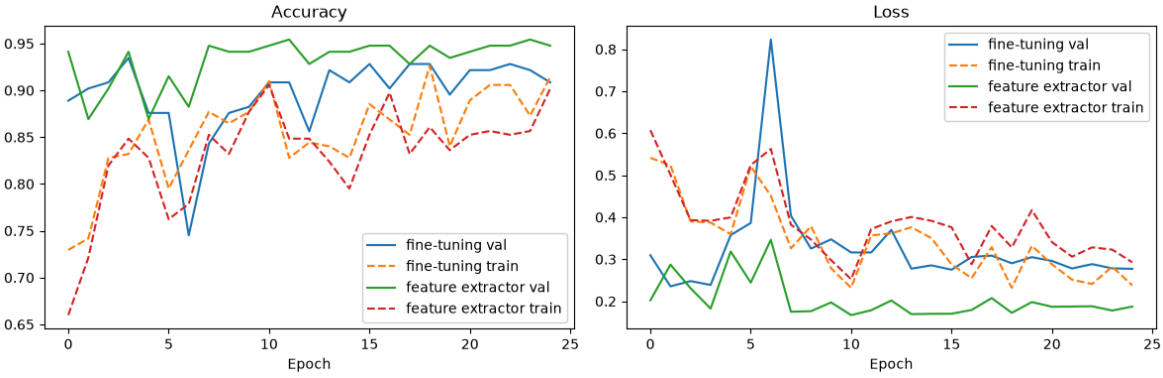
Q3(b): Feature extractor

The pretrained network is frozen and only the new final layer (1,026 parameters) is trained, so the network is used as a fixed feature extractor. The best validation accuracy was 95.42%, and the last epoch gave 94.8%.

In [30]:
```python
# Comparison table and curves for (a) and (b)
print(f"{'':20s} {'best val':>9s} {'last val':>9s} {'last train':>11s} {'s/epoch':>8s} {'trainable params':>17s}")
for name, h, best, tp in [("Q3(a) fine-tuning", hist_ft, best_ft, trainable_ft),
                          ("Q3(b) feat. extractor", hist_fe, best_fe, trainable_fe)]:
    print(f"{name:20s} {100*best:8.2f}% {100*h['val_acc'][-1]:8.2f}% {100*h['train_acc'][-1]:10.2f}% "
          f"{np.mean(h['time']):8.1f} {tp:17,d}")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
for name, h in [("fine-tuning", hist_ft), ("feature extractor", hist_fe)]:
    ax[0].plot(h['val_acc'], label=f"{name} val"); ax[0].plot(h['train_acc'], '--', label=f"{name} train")
    ax[1].plot(h['val_loss'], label=f"{name} val"); ax[1].plot(h['train_loss'], '--', label=f"{name} train")
ax[0].set_title("Accuracy"); ax[1].set_title("Loss")
for a in ax: a.set_xlabel("Epoch"); a.legend()
plt.tight_layout(); plt.show()
```

```
                      best val  last val  last train  s/epoch  trainable params
Q3(a) fine-tuning       93.46%    90.85%      91.39%     18.1        11,177,538
Q3(b) feat. extractor   95.42%    94.77%      90.16%     15.9             1,026
```



Q3: Comparison and discussion

	Best val acc	Last-epoch val acc	Trainable parameters	Time per epoch
(a) Fine-tuning	93.46%	90.85%	11,177,538	18.1s
(b) Feature extractor	95.42%	94.77%	1,026	15.9s

The feature extractor did at least as well as fine-tuning here. Its best validation accuracy was 95.42% (146 of 153 images) against 93.46% (143 of 153), a gap of 3 images. On the last epoch the gap is larger: 94.77% against 90.85%, or 6 images. Each image is worth 0.65 points and this is a single run, so the feature extractor is competitive and probably slightly better, but not clearly better. A likely reason is the very small dataset: updating 11 million weights from 244 images can disturb good pretrained features, whereas training only the last layer cannot. Features learned on ImageNet (edges, textures, shapes, animal parts) already separate ants from bees well.

The feature extractor trains far fewer parameters (1,026 vs 11.2 million) and does no backward pass through the backbone, but its time per epoch was only about 12% lower (15.9 s vs 18.1 s). With a small dataset and batch size 4, much of the time is probably spent on loading and augmenting images and on the forward pass, which both approaches share. Epoch times also varied between about 13 and 20 seconds, so this difference should not be over-interpreted. With a larger dataset, or one very different from ImageNet, fine-tuning would be expected to gain more.

For the feature extractor, training accuracy stays below validation accuracy at every epoch (about 90% vs 95% at the end). For fine-tuning the two are about equal in later epochs. For the feature extractor, the gap exists because training images are randomly cropped and flipped, which makes them harder than the fixed center-cropped validation images. Also, the best epoch is chosen using the same validation images that the accuracy is reported on, so "best val acc" is slightly optimistic. The last-epoch value is a fairer estimate.